

1

Examples can vary.

(a) Natural number:

$$\boxed{5}$$

(b) Integer that is not natural:

$$\boxed{-2}$$

(c) Rational number that is not an integer:

$$\boxed{\frac{1}{2}}$$

(d) Irrational number:

$$\boxed{\sqrt{2}}$$

2

(a)

$$3.14 \times 10^7 = 31,400,000$$

(b)

$$2.764 \times 10^6 = 2,764,000$$

(c)

$$2.61 \times 10^{-3} = 0.00261$$

(d)

$$9.999 \times 10^{-2} = 0.09999$$

3

$$x^3 + 10x^2 + 25x$$

Factor out x :

$$x(x^2 + 10x + 25)$$

$$x(x + 5)^2$$

$$\boxed{x(x + 5)^2}$$

4

$$\frac{9}{2 - \sqrt{5}}$$

Multiply by the conjugate:

$$\frac{9}{2 - \sqrt{5}} \cdot \frac{2 + \sqrt{5}}{2 + \sqrt{5}}$$

$$= \frac{9(2 + \sqrt{5})}{4 - 5}$$

$$= \frac{9(2 + \sqrt{5})}{-1}$$

$$\boxed{-18 - 9\sqrt{5}}$$

5

Let width be w . Then length is $w + 5$. The diagonal is 145.

$$w^2 + (w + 5)^2 = 145^2$$

$$w^2 + w^2 + 10w + 25 = 21025$$

$$2w^2 + 10w - 21000 = 0$$

Divide by 2:

$$w^2 + 5w - 10500 = 0$$

Factor:

$$(w - 100)(w + 105) = 0$$

Since width must be positive:

$$w = 100$$

$$\text{length} = 100 + 5 = 105$$

width = 100 m, length = 105 m

6

(a)

$$9x^2 - 25y^2 = 225$$

For x-intercepts, set $y = 0$:

$$9x^2 = 225$$

$$x^2 = 25$$

$$x = \pm 5$$

x-intercepts:

$(-5, 0)$	,	$(5, 0)$
-----------	---	----------

For y-intercepts, set $x = 0$:

$$-25y^2 = 225$$

$$y^2 = -9$$

No real y-intercept.

DNE

(b)

$$y^2 - 2xy + 5x = 1$$

For x-intercept, set $y = 0$:

$$5x = 1$$

$$x = \frac{1}{5}$$

$\left(\frac{1}{5}, 0\right)$

For y-intercept, set $x = 0$:

$$y^2 = 1$$

$$y = \pm 1$$

$$\boxed{(0, 1)}, \quad \boxed{(0, -1)}$$

7

" P varies inversely as S " means:

$$P = \frac{k}{S}$$

$$\boxed{P = \frac{k}{S}}$$

8

$$f(x) = 3x - 2$$

$$g(x) = 4 - x^2$$

(a)

$$(f \circ f)(-1) = f(f(-1))$$

$$f(-1) = 3(-1) - 2 = -5$$

$$f(-5) = 3(-5) - 2 = -17$$

$$\boxed{-17}$$

(b)

$$(g \circ g)(-1) = g(g(-1))$$

$$g(-1) = 4 - (-1)^2 = 3$$

$$g(3) = 4 - 3^2 = 4 - 9 = -5$$

$$\boxed{-5}$$

9

$$f(x) = -x + 5, \quad -5 \leq x \leq 5$$

Evaluate the given values:

$$f(-5) = -(-5) + 5 = 10$$

$$f(-4) = -(-4) + 5 = 9$$

$$f(0) = 5$$

$$f(1) = -1 + 5 = 4$$

$$f(4) = -4 + 5 = 1$$

$$f(5) = -5 + 5 = 0$$

Table:

10, 9, 5, 4, 1, 0

The graph is a line segment from $(-5, 10)$ to $(5, 0)$.

10

$$f(x) = |3x - 3|$$

Evaluate the table:

$$f(-5) = |3(-5) - 3| = |-18| = 18$$

$$f(-2) = |3(-2) - 3| = |-9| = 9$$

$$f(0) = |-3| = 3$$

$$f(1) = |0| = 0$$

$$f(2) = |3| = 3$$

$$f(5) = |12| = 12$$

Table:

18, 9, 3, 0, 3, 12

The graph is a V-shape with vertex at $(1, 0)$. The correct graph is the upward-opening V with vertex on the x-axis at $x = 1$.

11

$$r(x) = \frac{3x - 4}{x^2 - 16}$$

Factor denominator:

$$x^2 - 16 = (x - 4)(x + 4)$$

Vertical asymptotes occur where the denominator is zero and no factor cancels.

$$x = 4, \quad x = -4$$

So vertical asymptotes:

$$x = 4, x = -4$$

For the horizontal asymptote, compare degrees. The numerator has degree 1 and denominator has degree 2, so the denominator grows faster.

$$y = 0$$

12

$$\log(x) = -5$$

Assuming common logarithm:

$$x = 10^{-5}$$

$$x = 10^{-5}$$

or

$$x = 0.00001$$

13

$$\sin(\sin^{-1}(\frac{1}{8}))$$

Since $\sin$ and $\sin^{-1}$ undo each other for values in $[-1, 1]$:

$$\frac{1}{8}$$

14

Use the Law of Sines.

Side $AB = 80.2$ is opposite angle $C = 66^\circ$. Side $AC = 56.3$ is opposite angle $B = \theta$.

$$\frac{\sin \theta}{56.3} = \frac{\sin 66^\circ}{80.2}$$

$$\sin \theta = \frac{56.3 \sin 66^\circ}{80.2}$$

$$\sin \theta \approx \frac{56.3(0.9135)}{80.2}$$

$$\sin \theta \approx 0.6414$$

$$\theta = \sin^{-1}(0.6414)$$

$$\theta \approx 39.9^\circ$$

$$\boxed{39.9^\circ}$$

15

$$y = 6x^2$$

Use polar substitutions:

$$x = r \cos \theta$$

$$y = r \sin \theta$$

Substitute:

$$r \sin \theta = 6(r \cos \theta)^2$$

$$r \sin \theta = 6r^2 \cos^2 \theta$$

Divide by r , assuming $r \neq 0$:

$$\sin \theta = 6r \cos^2 \theta$$

Solve for r :

$$r = \frac{\sin \theta}{6 \cos^2 \theta}$$

16

Length of major axis is 8, so:

$$2a = 8$$

$$a = 4$$

Length of minor axis is 6, so:

$$2b = 6$$

$$b = 3$$

The foci are on the x-axis, so the major axis is horizontal.

Standard form:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

$$\boxed{\frac{x^2}{16} + \frac{y^2}{9} = 1}$$

17

$$(1 - x)^6$$

Use the binomial coefficients for power 6:

1, 6, 15, 20, 15, 6, 1

$$(1 - x)^6 = 1 - 6x + 15x^2 - 20x^3 + 15x^4 - 6x^5 + x^6$$

$$\boxed{1 - 6x + 15x^2 - 20x^3 + 15x^4 - 6x^5 + x^6}$$





 Sources